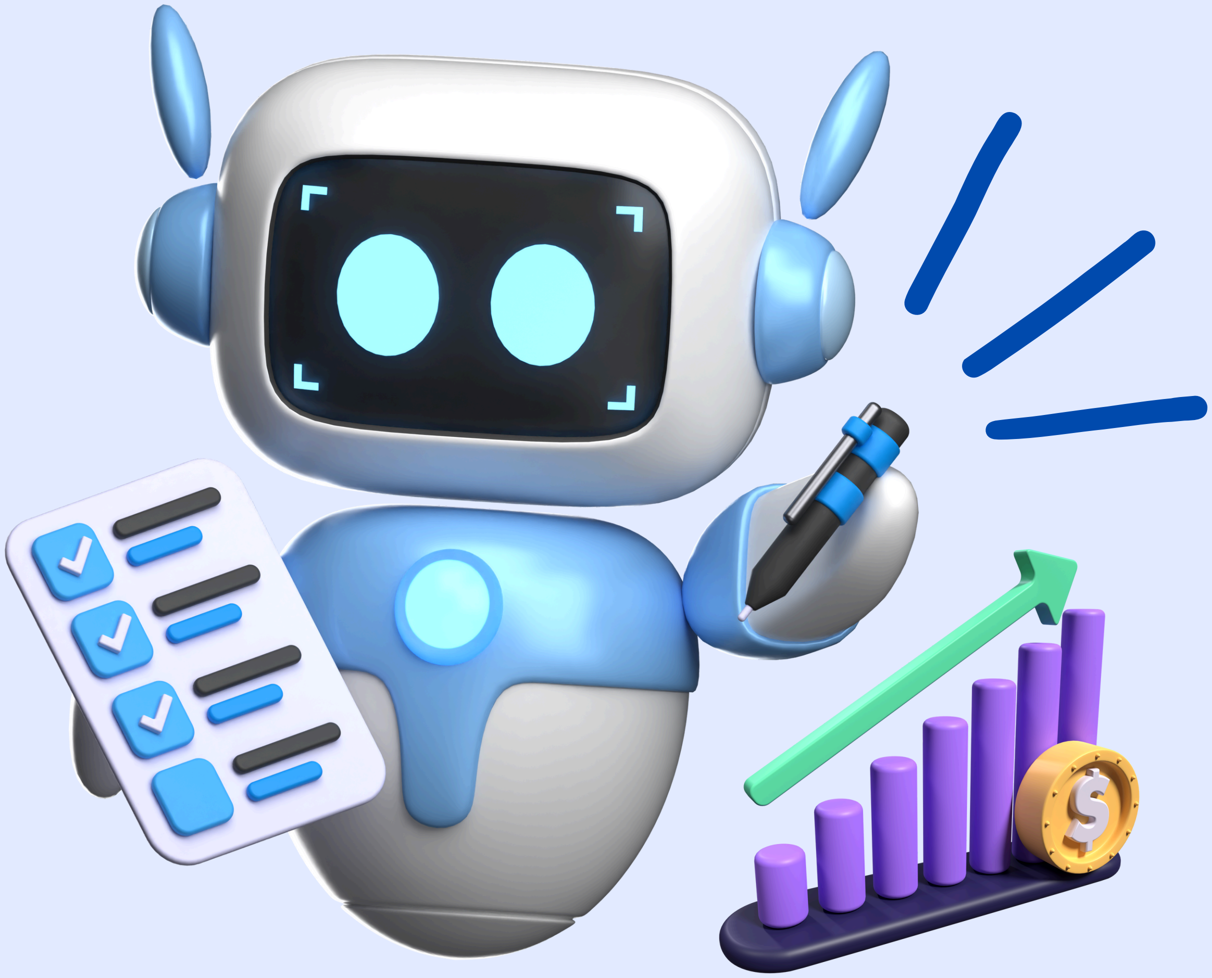
How to use AI + Python for FP&A Forecasting

with CODE!

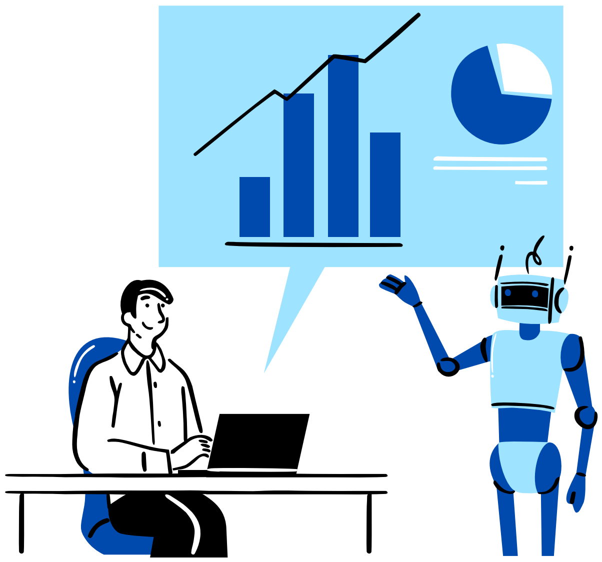


CHRISTIAN MARTINEZ
SENIOR FINANCE TRANSFORMATION MANAGER

What are the tools we will use:

We will use:

- **ChatGPT:** to help generate and improve code
- **Python**: to build dynamic models
- **Google Colab**: to run code without installing anything (you can also use Python in Excel or Visual Studio)
- **Excel:** to store and structure your financial data



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Step 1: Format Your Financial Data

Your Excel should have:

- Monthly financials (P&L format: Revenue, COGS, OpEx, EBITDA, etc.)
- Driver data like: Customer Base, ARPU, Churn Rate, New Customers

Most of us store data horizontally in Excel (like this →):

	2023 January	2023 February	2023 March	2023 April	2023 May	2023 June
Revenue	$97,696.81	$113,497.89	$115,053.33	$109,197.90	$129,024.17	$128,986.17
COGS	$17,938.50	$22,175.26	$20,668.38	$21,364.18	$19,844.25	$24,277.65
R&D	$29,215.78	$27,072.97	$30,592.24	$30,522.11	$30,010.23	$29,530.83
Sales & Marketing	$42,857.06	$41,879.83	$41,977.58	$37,726.53	$43,506.99	$36,495.37
G&A	$17,588.78	$20,276.95	$20,389.82	$21,172.73	$18,144.57	$18,019.32
EBITDA	**-$9,903.31**	**$2,092.88**	**$1,425.31**	**-$1,587.66**	**$17,518.14**	**$20,663.00**
D&A	$3,411.03	$3,948.40	$2,877.31	$2,623.13	$2,555.24	$2,592.09
EBIT	-$13,314.34	-$1,855.52	-$1,452.00	-$4,210.79	$14,962.89	$18,070.91
Interest	$1,000.00	$1,000.00	$1,000.00	$1,000.00	$1,000.00	$1,000.00
Tax	$0.00	$0.00	$0.00	$0.00	$3,740.72	$4,517.73
Net Income	**-$17,725.37**	**-$6,803.92**	**-$5,329.30**	**-$7,833.92**	**$7,666.93**	**$9,961.09**

Customer Base	1000	1078	1131.1	1197.545	1269.66775	1315.18436
New Customers	116	128	107	123	132	109
Churn Rate	0.05	0.05	0.05	0.05	0.05	0.05
ARPU	97.7	105.29	101.72	91.18	101.62	98.07

But in Python, you want it vertical
(one row per month, like this →)

Date	ustomer Ba	w Custom	Churn Rate	ARPU	Revenue	COGS
2023-01-01 00:00:00	1000	116	0.05	97.7	97696.81	17938.5
2023-02-01 00:00:00	1078	128	0.05	105.29	113497.9	22175.26
2023-03-01 00:00:00	1131.1	107	0.05	101.72	115053.3	20668.38
2023-04-01 00:00:00	1197.545	123	0.05	91.18	109197.9	21364.18
2023-05-01 00:00:00	1269.668	132	0.05	101.62	129024.2	19844.25
2023-06-01 00:00:00	1315.184	109	0.05	98.07	128986.2	24277.65
2023-07-01 00:00:00	1362.425	113	0.05	96.62	131631.2	23808.2
2023-08-01 00:00:00	1412.304	118	0.05	103.06	145549.8	25763.75
2023-09-01 00:00:00	1465.689	124	0.05	105.15	154124.5	27883.83
2023-10-01 00:00:00	1505.404	113	0.05	104.66	157550.2	25227.62
2023-11-01 00:00:00	1549.134	119	0.05	95.8	148413.1	26388.34
2023-12-01 00:00:00	1580.677	109	0.05	98.45	155623.9	28568.06
2024-01-01 00:00:00	1624.643	123	0.05	101.66	165155.3	32168.77
2024-02-01 00:00:00	1666.411	123	0.05	104.88	174769.4	30552.72
2024-03-01 00:00:00	1696.091	113	0.05	97.6	165545.5	28459.76
2024-04-01 00:00:00	1737.286	126	0.05	99.07	172115.9	30117.26
2024-05-01 00:00:00	1755.422	105	0.05	94.47	165831.8	31367.75
2024-06-01 00:00:00	1779.651	112	0.05	94.02	167320.9	30667.84
2024-07-01 00:00:00	1804.668	114	0.05	104.06	187798.5	32808.85
2024-08-01 00:00:00	1835.435	121	0.05	106.78	195989.9	36284.14
2024-09-01 00:00:00	1866.663	123	0.05	99.64	185994.2	33659.52
2024-10-01 00:00:00	1875.33	102	0.05	105.02	196942.8	37357.38
2024-11-01 00:00:00	1875.563	94	0.05	101.81	190947.7	33030.03
2024-12-01 00:00:00	1897.785	116	0.05	96.77	183657	32456.49

You can easily convert this in Excel using ChatGPT, Power Query or Python!

Tell me in the comments if you want this code

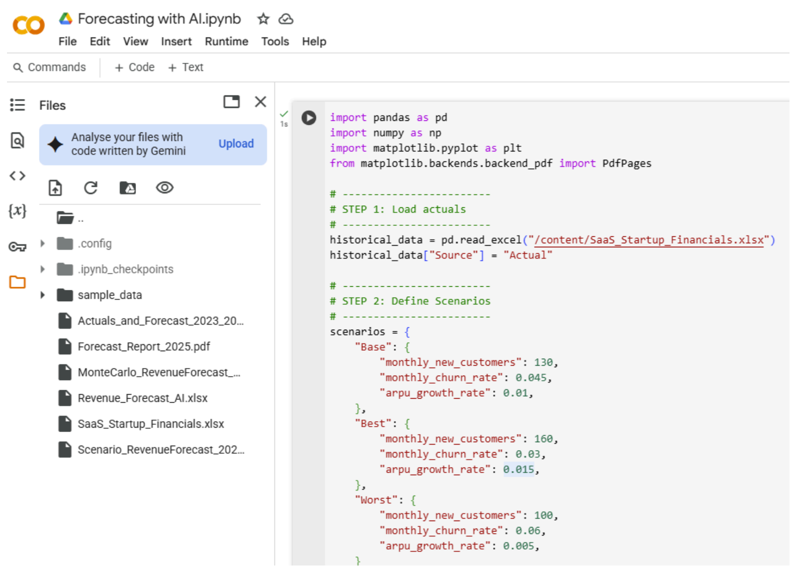
BY CHRISTIAN MARTINEZ

Step 2: Use ChatGPT + Python to Forecast

Once your data is structured, you can run this AI-assisted P&L forecasting engine.

My code includes:

- Base, Best, and Worst scenarios
- Dynamic drivers (ARPU growth, churn, new customers)
- Full calculation down to Net Income
- Auto-generated charts
- Outputs a clean PDF report




```python
# Common assumptions
base_arpu = historical_data["ARPU"].iloc[-1]
starting_customers = historical_data["Customer Base"].iloc[-1]
cogs_pct = 0.18
cogs_noise = 0.01
r_and_d_base = 32000
r_and_d_growth = 0.01
s_and_m_base = 42000
s_and_m_growth = 0.015
g_and_a_base = 21000
g_and_a_growth = 0.01
da_base = 3200
interest = 1000
tax_rate = 0.25
```

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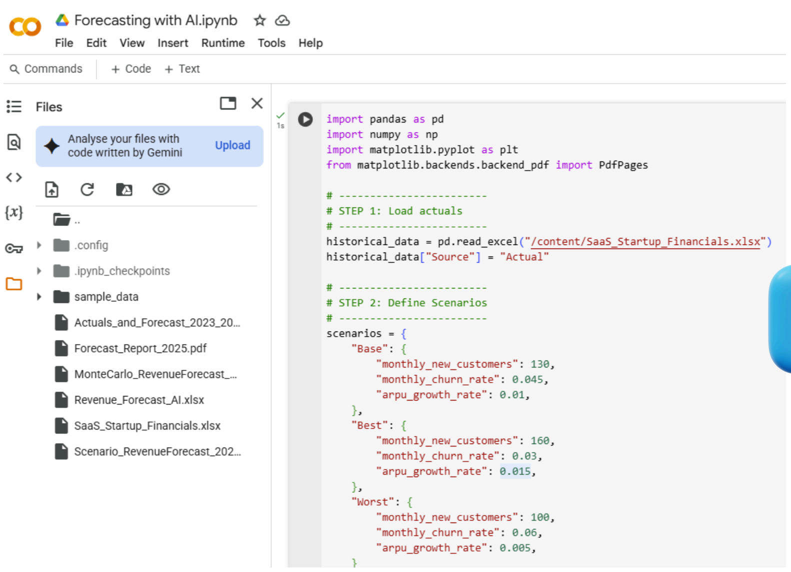
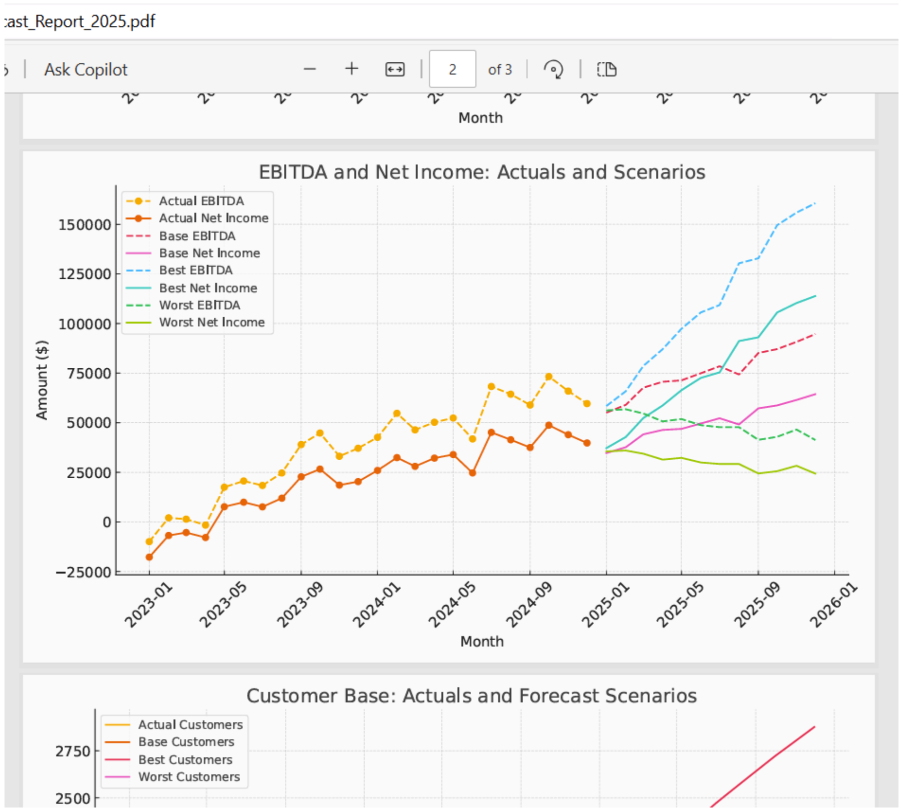
Step 3: Generate PDF Reports with Charts

Copy my code that gives you a ready-to-present PDF with:

- 📈 Revenue forecast across scenarios
- 💵 Net Income + EBITDA lines
- 📉 Customer base projections





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Step 4: Get the Forecast in Excel

After generating your forecast in Python (with scenarios, drivers, P&L lines), you can save it as an Excel file with my code:

```python
# ----------------------------
# STEP 4: Combine & Export
# ----------------------------
combined_df = pd.concat([historical_data, forecast], ignore_index=True)
combined_df.to_excel("Actuals_and_Forecast_2023_2025.xlsx", index=False)

print("Excel file created: Actuals_and_Forecast_2023_2025.xlsx")
```

```
Excel file created: Actuals_and_Forecast_2023_2025.xlsx
```

This will save the full forecast — including:

- Actuals from your Excel file
- Base / Best / Worst scenarios
- Revenue, COGS, R&D, S&M, G&A
- EBITDA, EBIT, Net Income
- Customer base & ARPU

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More Ideas to Try with AI + Python in Forecasting

You've built your first FP&A forecast using Python and AI.

You've got:

✅ Dynamic drivers
✅ Scenario planning
✅ Beautiful PDF reports
✅ Excel output

So… what's next?

Here are 3 more practical ideas to take your forecasting game to the next level 👇

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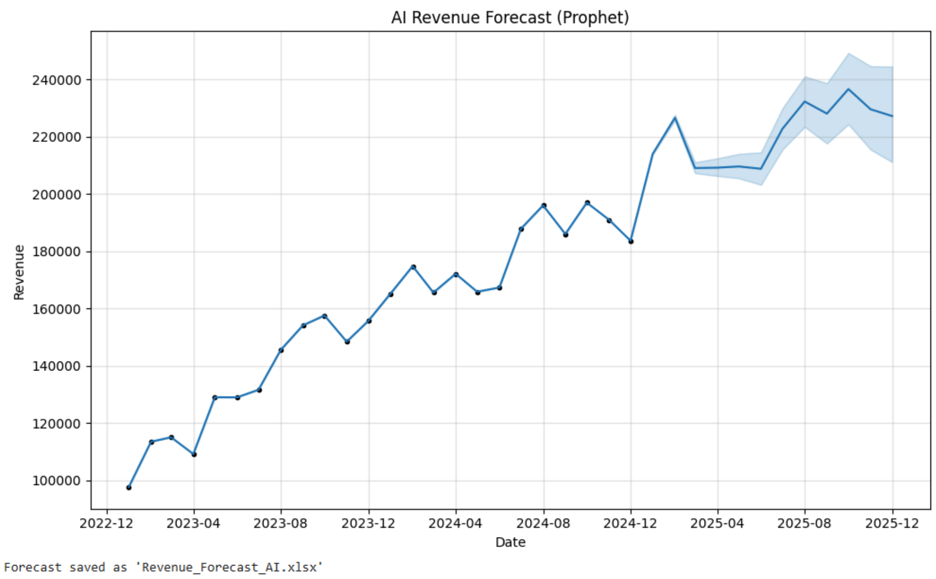
1) Trend Detection & Seasonality (Time Series Modeling)

AI can detect complex patterns in historical data:

- Machine learning models (e.g., XGBoost, Random Forest) can forecast revenue, churn, and cost behavior based on time and other drivers.

- Time series models (like Facebook Prophet, ARIMA, or LSTM) handle:
  - Seasonality (e.g., annual subscription renewals)
  - Holidays/events
  - Multi-year trends



📈 Example: Predict churn or monthly revenue using historical patterns + features like marketing spend, customer segments, or macro data.

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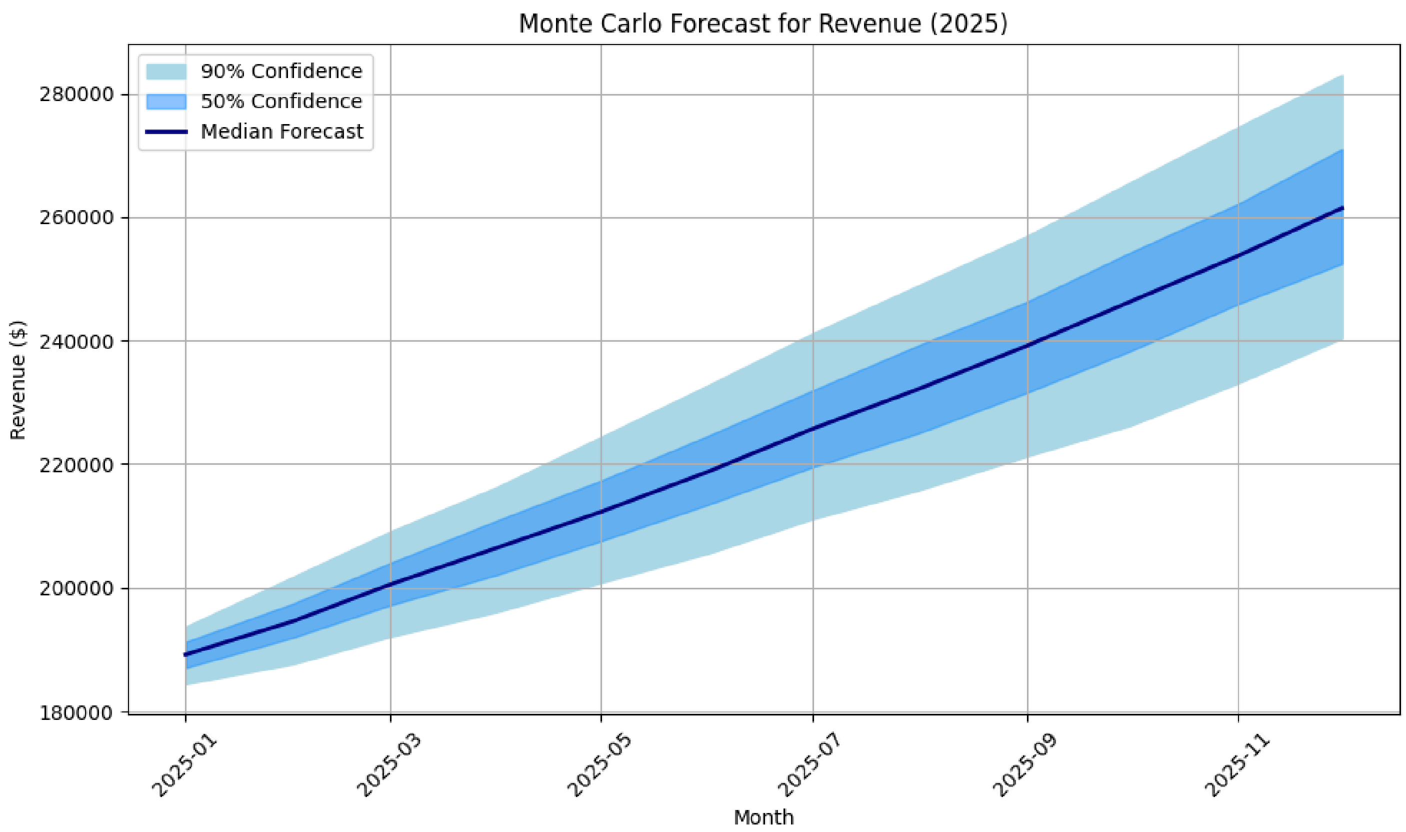
2) Scenario Simulation (Monte Carlo / Probabilistic Forecasting)

AI helps simulate thousands of possible futures, not just a “Base Case”:

- Add uncertainty to assumptions like churn rate, ARPU growth, or marketing ROI.
- Use Monte Carlo simulations to generate a range of outcomes and confidence intervals.

🧪 Example: “If churn varies between 4–7%, what’s the likely range of Net Income?”



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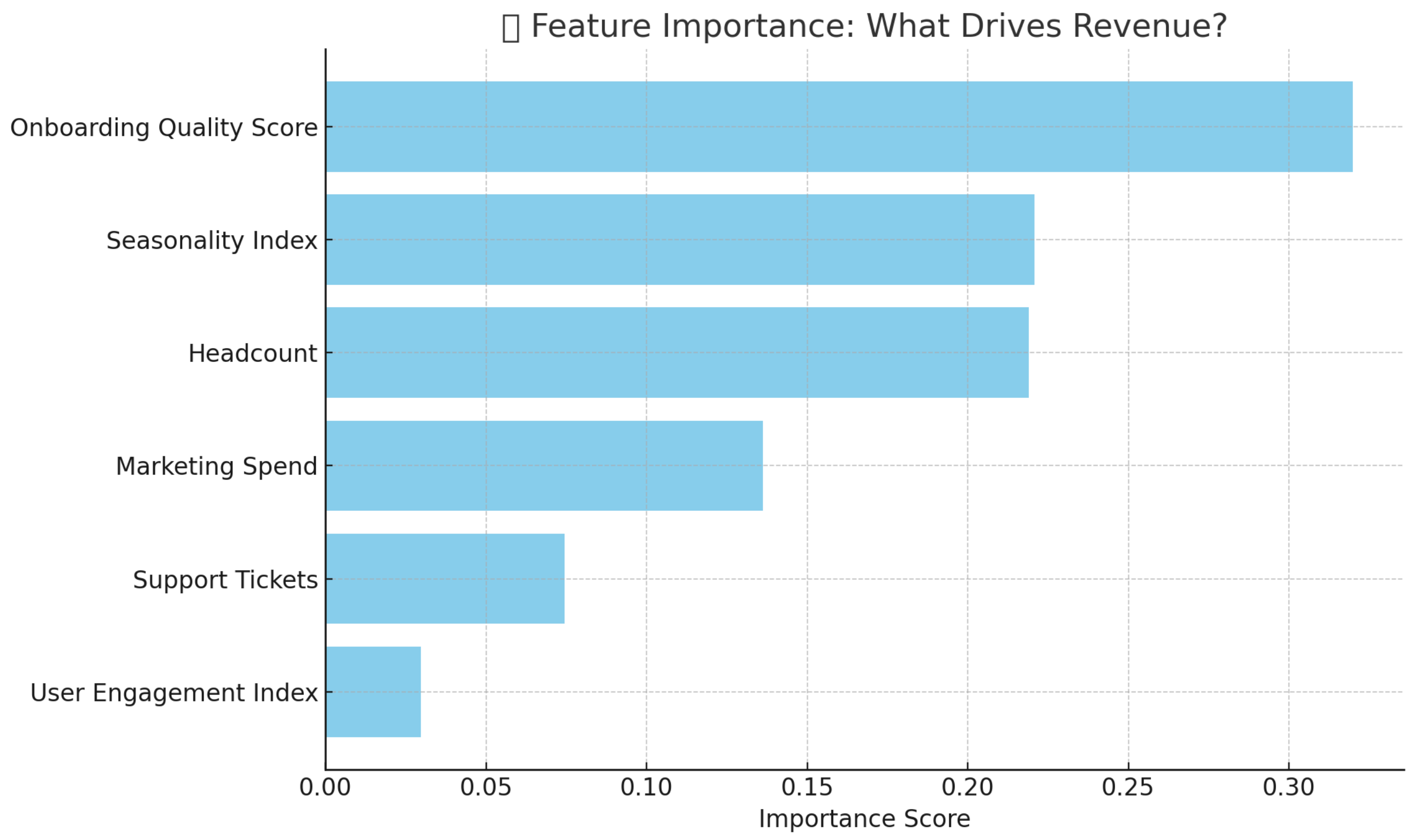
3) Dynamic Driver Identification (Feature Importance)

Using ML models, AI can:

- Discover what really drives revenue or cost (e.g., headcount, ad spend, seasonality)
- Prioritize levers with the most impact

📊 Example: AI may discover that onboarding experience quality correlates with long-term ARPU more than marketing spend.



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Here is a Summary:

Step 1: Format Your Financial Data

Step 2: Use ChatGPT + Python to Forecast

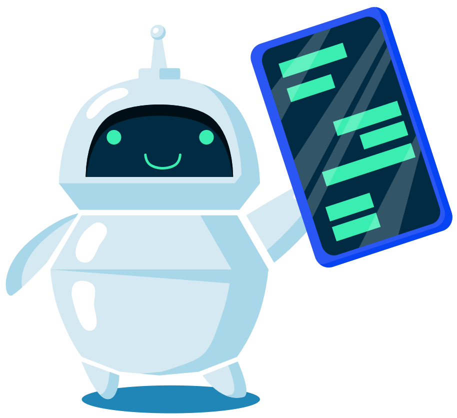
Step 3: Generate PDF Reports with Charts

Step 4: Get the Forecast in Excel

More ideas:

- Trend Detection & Seasonality (Time Series Modeling)
- Scenario Simulation (Monte Carlo / Probabilistic Forecasting)
- Dynamic Driver Identification (Feature Importance)

Christian Martinez



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